Optimization

Francesca Maggioni

Lesson 6

Department of Management, Information and Production Engineering
University of Bergamo

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Can We Go Even Faster?

- So far: Error decreases with $1/\sqrt{t}$, or $1/T$...
- Could it decrease exponentially in T?
- On $f(x) := x^2$. Stepsize $\gamma := 1/2$ (f is $L = 2$-smooth):

$$x_{t+1} := x_t - \frac{1}{2}\nabla f(x_t) = x_t - x_t = 0$$

  converged in one step!
- Same for $f(x) := x^2$. Stepsize $\gamma := 1/4$ (f is $L = 4$-smooth):

$$x_{t+1} := x_t - \frac{1}{4}\nabla f(x_t) = x_t - \frac{x_t}{2} = \frac{x_t}{2}$$

  so $f(x_t) = f(\frac{x_0}{2^t}) = \frac{1}{2^{2t}}x_0^2$. Exponential in t.

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Strongly Convex Functions

Definition

Let $f : dom(f) \to \mathbb{R}$ be a differentiable function, $X \subseteq dom(f)$ convex and $\mu \in \mathbb{R}_+, \mu > 0$. Function f is called **strongly convex** *(with parameter μ) over X if:*

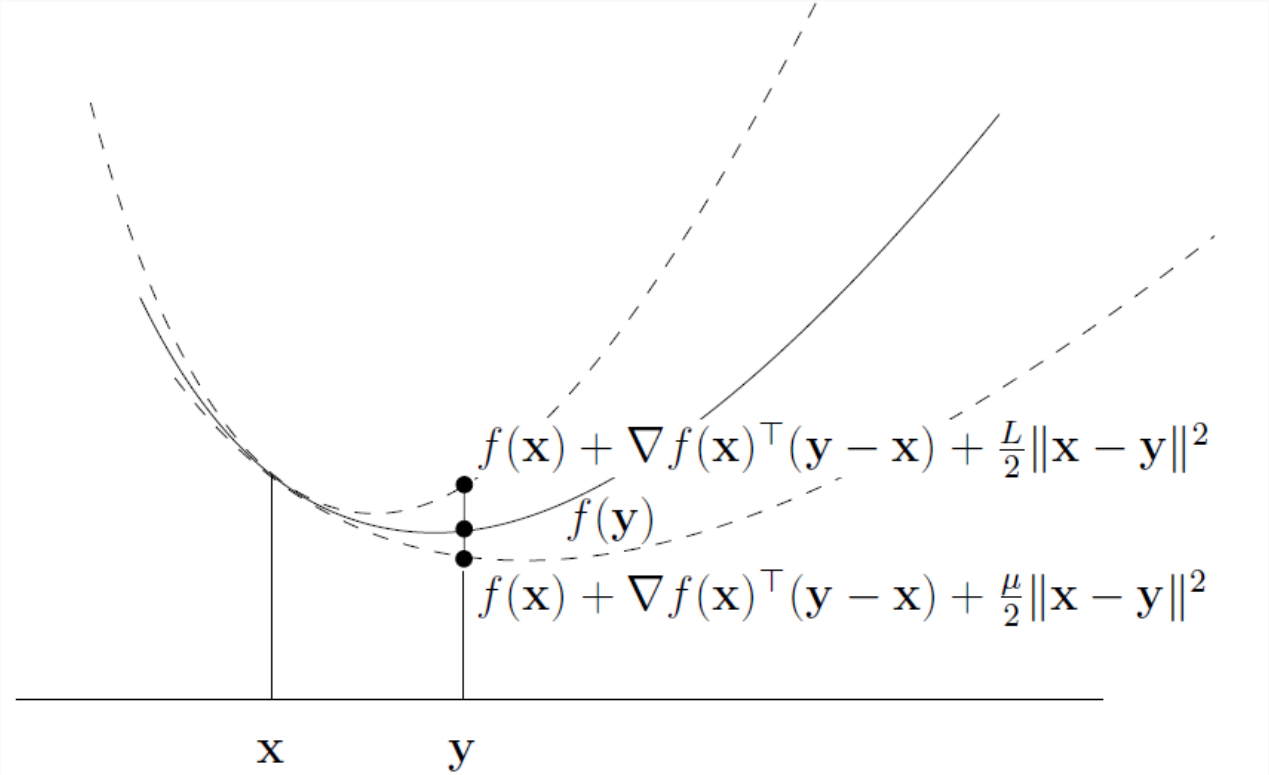
$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} ||\mathbf{x} - \mathbf{y}||^2, \quad \forall\, \mathbf{x}, \mathbf{y} \in X.$$

Lemma

If f is strongly convex with parameter $\mu > 0$, then f is strictly convex and has a unique global minimum.

Strongly Convex Functions

Strong convexity: For any $\mathbf{x}$, the graph of f is above a not too flat tangent paraboloid at $(\mathbf{x}, f(\mathbf{x}))$:



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Smooth and Strongly Convex Functions: $\mathcal{O}(\log(1/\varepsilon))$ Steps

Want to show: $\lim_{t\to\infty} \mathbf{x}_t = \mathbf{x}^*$.

Vanilla Analysis:

$$\nabla f(\mathbf{x}_t)^\top (\mathbf{x}_t - \mathbf{x}^*) = \frac{\gamma}{2}||\nabla f(\mathbf{x}_t)||^2 + \frac{1}{2\gamma}\left(||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{x}_{t+1} - \mathbf{x}^*||^2\right).$$

Now use stronger lower bound on left hand side, coming from strong convexity:

$$\nabla f(\mathbf{x}_t)^\top (\mathbf{x}_t - \mathbf{x}^*) \geq f(\mathbf{x}_t) - f(\mathbf{x}^*) + \frac{\mu}{2}||\mathbf{x}_t - \mathbf{x}^*||^2.$$

Putting it together:

$$f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \frac{1}{2\gamma}\left(\gamma^2||\nabla f(\mathbf{x}_t)||^2 + ||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{x}_{t+1} - \mathbf{x}^*||^2\right) - \frac{\mu}{2}||\mathbf{x}_t - \mathbf{x}^*||^2.$$

Rewriting:

$$||\mathbf{x}_{t+1} - \mathbf{x}^*||^2 \leq 2\gamma\left(f(\mathbf{x}^*) - f(\mathbf{x}_t)\right) + \gamma^2||\nabla f(\mathbf{x}_t)||^2 + (1 - \mu\gamma)||\mathbf{x}_t - \mathbf{x}^*||^2.$$



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Smooth and Strongly Convex Functions: $\mathcal{O}(\log(1/\varepsilon))$ Steps

Squared distance to $\mathbf{x}^*$ goes down by a constant factor, up to some noise.

Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable with a global minimum $\mathbf{x}^*$; suppose that f is smooth with parameter L and strongly convex with parameter $\mu > 0$. Choosing $\gamma := \frac{1}{L}$, gradient descent with arbitrary $\mathbf{x}_0$ satisfies the following two properties.

1. *Squared distances to $\mathbf{x}^*$ are geometrically decreasing:*

$$||\mathbf{x}_{t+1} - \mathbf{x}^*||^2 \leq \left(1 - \frac{\mu}{L}\right) ||\mathbf{x}_t - \mathbf{x}^*||^2, \quad t \geq 0. \qquad (\star)$$

2. *The absolute error after T iterations is exponentially small in T:*

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{L}{2} \left(1 - \frac{\mu}{L}\right)^T ||\mathbf{x}_0 - \mathbf{x}^*||^2, \quad T > 0. \qquad (\star\star)$$

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Smooth and Strongly Convex Functions: $\mathcal{O}(\log(1/\varepsilon))$ Steps

Proof of $(\star)$.

Bounding the noise:

$$\begin{aligned}
2\gamma\left(f(\mathbf{x}^*) - f(\mathbf{x}_t)\right) + \gamma^2 ||\nabla f(\mathbf{x}_t)||^2 &= \frac{2}{L}\left(f(\mathbf{x}^*) - f(\mathbf{x}_t)\right) + \frac{1}{L^2}||\nabla f(\mathbf{x}_t)||^2 \\
&\leq \frac{2}{L}\left(f(\mathbf{x}_{t+1}) - f(\mathbf{x}_t)\right) + \frac{1}{L^2}||\nabla f(\mathbf{x}_t)||^2 \\
&\leq -\frac{1}{L^2}||\nabla f(\mathbf{x}_t)||^2 + \frac{1}{L^2}||\nabla f(\mathbf{x}_t)||^2 = 0.
\end{aligned}$$

Hence, the noise is non-positive, and we get $(\star)$:

$$||\mathbf{x}_{t+1} - \mathbf{x}^*||^2 \leq (1 - \mu\gamma)||\mathbf{x}_t - \mathbf{x}^*||^2 = \left(1 - \frac{\mu}{L}\right)||\mathbf{x}_t - \mathbf{x}^*||^2.$$

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Smooth and Strongly Convex Functions: $\mathcal{O}(\log(1/\varepsilon))$ Steps

Proof of 2.

From $(\star)$:

$$||\mathbf{x}_T - \mathbf{x}^*||^2 \leq \left(1 - \frac{\mu}{L}\right)^T ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

Smoothness together with $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \nabla f(\mathbf{x}^*)^\top (\mathbf{x}_T - \mathbf{x}^*) + \frac{L}{2}||\mathbf{x}_T - \mathbf{x}^*||^2.$$

Putting it together:

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{L}{2}||\mathbf{x}_T - \mathbf{x}^*||^2 \leq \frac{L}{2}\left(1 - \frac{\mu}{L}\right)^T ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

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Smooth and Strongly Convex Functions: $\mathcal{O}(\log(1/\varepsilon))$ Steps

$R^2 := ||\mathbf{x}_0 - \mathbf{x}^*||^2$.

$$T \geq \frac{L}{\mu} \ln\left(\frac{R^2 L}{2\varepsilon}\right) \Rightarrow \text{ error} \leq \frac{L}{2}\left(1 - \frac{\mu}{L}\right)^T R^2 \leq \varepsilon.$$

Conclusion: To reach absolute error at most ε, we only need $\mathcal{O}(\log(1/\varepsilon))$ iterations, *e.g.*:

- $\frac{L}{\mu} \ln(50 \cdot R^2 L)$ iterations for error 0.01 ...
- ... as opposed to $50 \cdot R^2 L$ in the smooth case.

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Projected Gradient Descent: Constrained Optimization

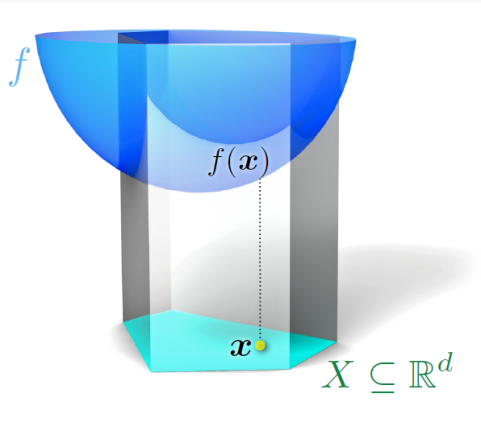
Constrained Optimization Problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ s.t. \quad & \mathbf{x} \in X. \end{aligned}$$

Solving Constrained Optimization Problems

(A) Projected Gradient Descent.

(B) Transform it into an unconstrained problem.



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Projected Gradient Descent: Constrained Optimization

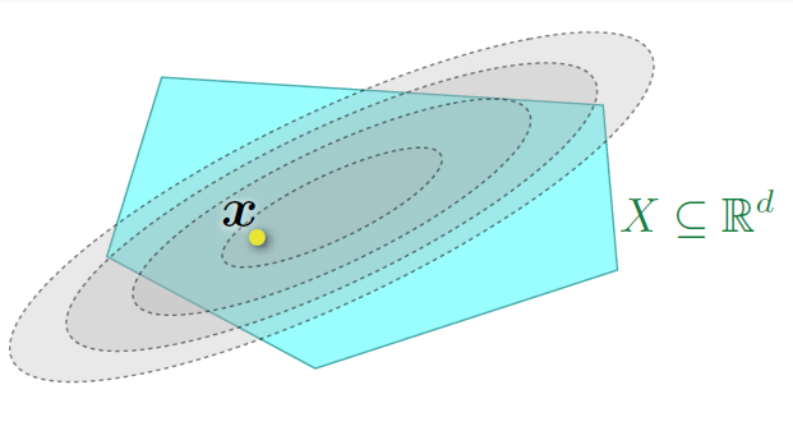
Constrained Optimization Problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ s.t. \quad & \mathbf{x} \in X. \end{aligned}$$

Solving Constrained Optimization Problems

(A) **Projected Gradient Descent $\Leftarrow$ HERE!**.

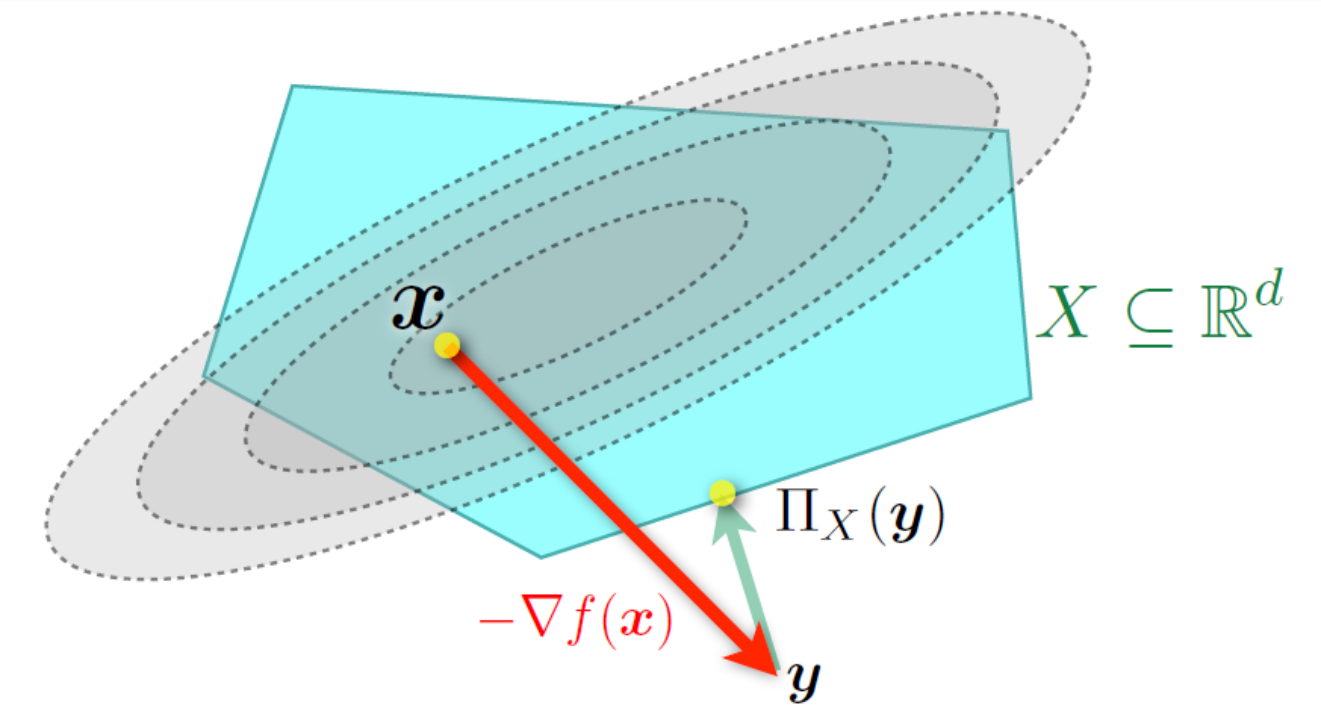
(B) Transform it into an unconstrained problem.



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Projected Gradient Descent

Idea: project onto X after every step: $\Pi_X(\mathbf{y}) := \text{argmin}_{\mathbf{x} \in X} ||\mathbf{x} - \mathbf{y}||$.



Projected gradient descent: $\mathbf{x}_{t+1} := \Pi_X \left[\mathbf{x}_t - \gamma \nabla f(\mathbf{x}_t)\right]$.

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The Algorithm

Projected gradient descent:

$$\mathbf{y}_{t+1} := \mathbf{x}_t - \gamma \nabla f(\mathbf{x}_t),$$
$$\mathbf{x}_{t+1} := \Pi_X(\mathbf{y}_{t+1}) := \text{argmin}_{\mathbf{x} \in X} ||\mathbf{x} - \mathbf{y}_{t+1}||^2.$$

for **timesteps** $t = 0, 1, \ldots,$ and **stepsize** $\gamma \geq 0$.

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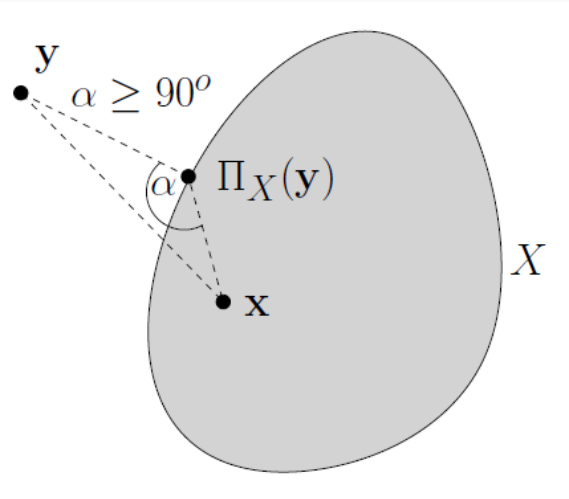
Properties of Projection

Fact

Let $X \subseteq \mathbb{R}^d$ be closed and convex, $\mathbf{x} \in X$, $\mathbf{y} \in \mathbb{R}^d$. Then:

(a) $(\mathbf{x} - \Pi_X(\mathbf{y}))^\top (\mathbf{y} - \Pi_X(\mathbf{y})) \leq 0.$

(b) $||\mathbf{x} - \Pi_X(\mathbf{y})||^2 + ||\mathbf{y} - \Pi_X(\mathbf{y})||^2 \leq ||\mathbf{x} - \mathbf{y}||^2.$



Properties of Projection

Proof of (a)

$\Pi_X(\mathbf{y})$ is minimizer of (differentiable) convex function $d_{\mathbf{y}}(\mathbf{x}) = ||\mathbf{x} - \mathbf{y}||^2$ over X. By first-order characterization of optimality

$$\begin{aligned} & 0 \leq & \nabla d_{\mathbf{y}}(\Pi_X(\mathbf{y}))^\top (\mathbf{x} - \Pi_X(\mathbf{y})) \\ & = & 2(\Pi_X(\mathbf{y}) - \mathbf{y})^\top (\mathbf{x} - \Pi_X(\mathbf{y})) \\ \iff & 0 \geq & 2(\mathbf{y} - \Pi_X(\mathbf{y}))^\top (\mathbf{x} - \Pi_X(\mathbf{y})) \\ \iff & 0 \geq & (\mathbf{x} - \Pi_X(\mathbf{y}))^\top (\mathbf{y} - \Pi_X(\mathbf{y})). \end{aligned}$$

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Properties of Projection

Proof of (b)

$$\mathbf{v} := (\mathbf{x} - \Pi_X(\mathbf{y})), \quad \mathbf{w} := (\mathbf{y} - \Pi_X(\mathbf{y})).$$

By (a):

$$\begin{aligned} 0 \geq \quad 2\mathbf{v}^\top \mathbf{w} = \quad & ||\mathbf{v}||^2 + ||\mathbf{w}||^2 - ||\mathbf{v} - \mathbf{w}||^2 \\ = \quad & ||\mathbf{x} - \Pi_X(\mathbf{y})||^2 + ||\mathbf{y} - \Pi_X(\mathbf{y})||^2 - ||\mathbf{x} - \mathbf{y}||^2. \end{aligned}$$

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Results for Projected Gradient Descent over Closed and Convex X

The **same** number of steps as gradient over $\mathbb{R}^d$!

- Lipschitz convex functions over X: $\mathcal{O}(1/\varepsilon^2)$ steps.
- Smooth convex functions over X: $\mathcal{O}(1/\varepsilon)$ steps.
- Smooth and strongly convex functions over X: $\mathcal{O}(\log(1/\varepsilon))$ steps.

We will adapt the previous proofs for gradient descent. BUT:

- Each step involves a projection onto X
- may or may not be efficient (in relevant cases, it is)...

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Lipschitz Convex Functions over X: $\mathcal{O}(1/\varepsilon^2)$ Steps

Assume that all gradients of f are bounded in norm over closed and convex X.

- Equivalent to f being Lipschitz over X.
- Many interesting functions are Lipschitz over bounded sets X.

Theorem (Same as the unconstrained one, but more useful)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable, $X \subseteq \mathbb{R}^d$ closed and convex, $\mathbf{x}^*$ a minimizer of f over X; furthermore, suppose that $||\mathbf{x}_0 - \mathbf{x}^*|| \leq R$ with $\mathbf{x}_0 \in X$ and $||\nabla f(\mathbf{x})|| \leq B$ for all $\mathbf{x} \in X$. Choosing the constant stepsize:

$$\gamma := \frac{R}{B\sqrt{T}}$$

projected *gradient descent yields*

$$\frac{1}{T}\sum_{t=0}^{T-1} f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \frac{RB}{\sqrt{T}}.$$

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Lipschitz Convex Functions over X: $\mathcal{O}(1/\varepsilon^2)$ Steps

Proof

- Replace $\mathbf{x}_{t+1}$ in the vanilla analysis with $\mathbf{y}_{t+1}$ (the unprojected gradient step):

$$\mathbf{g}_t^\top(\mathbf{x}_t - \mathbf{x}^*) = \frac{1}{2\gamma}\left(\gamma^2||\mathbf{g}_t||^2 + ||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{y}_{t+1} - \mathbf{x}^*||^2\right).$$

- Use Fact (b): $||\mathbf{x} - \Pi_X(\mathbf{y})||^2 + ||\mathbf{y} - \Pi_X(\mathbf{y})||^2 \leq ||\mathbf{x} - \mathbf{y}||^2$.
- With $\mathbf{x} = \mathbf{x}^*$, $\mathbf{y} = \mathbf{y}_{t+1}$ we have $\Pi_X(\mathbf{y}) = \mathbf{x}_{t+1}$, and hence:

$$||\mathbf{x}^* - \mathbf{x}_{t+1}||^2 \leq ||\mathbf{x}^* - \mathbf{y}_{t+1}||^2.$$

- We go back to the original vanilla analyis and continue from there as before:

$$\mathbf{g}_t^\top(\mathbf{x}_t - \mathbf{x}^*) \leq \frac{1}{2\gamma}\left(\gamma^2||\mathbf{g}_t||^2 + ||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{x}_{t+1} - \mathbf{x}^*||^2\right).$$

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Smooth Functions over X

Recall:

f is called smooth (with parameter L) over X if:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2}||\mathbf{x} - \mathbf{y}||^2, \quad \forall\, \mathbf{x}, \mathbf{y} \in X.$$

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Sufficient Decrease

Lemma

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and smooth with parameter L over X. Choosing stepsize

$$\gamma := \frac{1}{L},$$

projected gradient descent with arbitrary $\mathbf{x}_0 \in X$ satisfies

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \frac{1}{2L}||\nabla f(\mathbf{x}_t)||^2 + \frac{L}{2}||\mathbf{y}_{t+1} - \mathbf{x}_{t+1}||^2, \quad t \geq 0.$$

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Assignments

- Prove that the function $d_{\mathbf{y}} : \mathbb{R}^d \to \mathbb{R},\ \mathbf{x} \mapsto ||\mathbf{x} - \mathbf{y}||^2$ is strictly convex for any $\mathbf{y} \in \mathbb{R}^d$.
- Prove that $f(x) = x^4$ is smooth over $X = (-a; a)$ and determine a concrete smoothness parameter L.

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